Distribution-Aware Production Engineering
for Modern Optical and Photonic Systems:

From Gaussian Tolerancing through Distribution Identification
to Wigner-Negative Certification

Version History:
- **v0 (Mar 4, 2026):** Early LinkedIn post: "On My Upcoming Physics Colloquium Talk at NCU (posted date: March 4, 2026)"; thoughts on worked examples from basic to advanced.
- **v5 (Apr 7, 2026):** Preliminary technical document.

Abstract

Twentieth-century optical engineering treated fluctuations, tolerances, and noise as Gaussian by default. That assumption is the fixed point of coarse-graining at the limit where the Central Limit Theorem applies, and it served well for classical lens design and bulk radiometry. It does not survive the transition to silicon, thermal, and quantum photonics, where small mode counts, single-photon operation, multiplicative fabrication errors, and self-organized dynamics each break a different prerequisite of the CLT. The result is silent: yield overestimates at the foundry interface, NETD under-estimates in thermal radiometry, false-positive certifications of "single-photon" sources, biased active-alignment convergence, and unreliable acceptance decisions on the production floor.

This document presents **C8: Distribution-Aware Production Engineering**, the production-engineering counterpart to C3 (invariant-aware design optimization) in the Open-Source Engineering Optics Series. C8 is organized around *four inevitabilities* of non-Gaussianity — fundamental quantum-statistical (Bose–Einstein), compositional emergent (log-normal, Weibull), self-organized emergent (power-law, $1/f^\beta$, SOC avalanches), and quantum resource (Wigner negativity) — and audits the *five pillars* of any production-engineering program (design baseline, specification, assembly machines, inspection, capability reporting), each of which encodes a hidden Gaussian assumption that must be revised.

We provide six modules. Modules M1–M5 implement distribution identification, robust and distributionally-robust optimization, distribution-aware capability indices, Bayesian active alignment with two-distribution deconvolution, and tail-aware inspection with a quantum-tomography readout. Module M6 is a meta-audit layer: it returns bootstrap confidence intervals on every risk metric the upstream modules produce, translates the cost of choosing the wrong distribution family into a yield-regret matrix and a USD/year figure, and gives sample-size budgets as a function of tail heaviness. The toolkit is demonstrated on five test cases spanning silicon-photonic MZI yield, thermal-photonic NETD, heralded single-photon sources, fiber-to-PIC alignment, and self-organized phenomena (rogue waves and $1/f$ laser frequency noise). All eleven standalone simulation scripts are reproducible from synthetic data with seed 42 and follow the GAI file convention.

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List of Abbreviations

ANOVA	Analysis of Variance
B-life	Quantile-based reliability life
BIC	Bayesian Information Criterion
BO	Bayesian Optimization
CCDF	Complementary Cumulative Distribution Function
CLT	Central Limit Theorem
CPO	Co-Packaged Optics
Cpk	Process Capability Index
CVaR	Conditional Value-at-Risk
DRO	Distributionally Robust Optimization
DUT	Device Under Test
ER	Extinction Ratio
FoM	Figure of Merit
GP	Gaussian Process
KS	Kolmogorov–Smirnov (test)
LCB	Lower Confidence Bound
LSL/USL	Lower / Upper Specification Limit
MLE	Maximum Likelihood Estimation
MZI	Mach–Zehnder Interferometer
NETD	Noise-Equivalent Temperature Difference
NM	Nelder–Mead
PDF	Probability Density Function
PIC	Photonic Integrated Circuit
PSD	Power Spectral Density
QFI	Quantum Fisher Information
QKD	Quantum Key Distribution
RR	Repeatability & Reproducibility
RSS	Root-Sum-Square
SOC	Self-Organized Criticality
SOI	Silicon-on-Insulator
SPAD	Single-Photon Avalanche Diode
SPDC	Spontaneous Parametric Down-Conversion
SPRT	Sequential Probability Ratio Test
TPV	Thermophotovoltaic

Principal Symbols

α	Power-law / Pareto tail index
α_t	Avalanche-duration exponent
β	Weibull shape; PSD power-law exponent
Δx^*	Pre-bias offset $x_{\mathrm{CVaR}} - x_{\mathrm{nom}}$
η	Coupling efficiency / transmission
F	Fano factor $\mathrm{Var}(n)/\langle n\rangle$
$f(\mathbf{x}, \xi)$	Performance function with perturbation
$\langle n\rangle$	Mean photon occupation per mode
$\mathcal{N}(\rho)$	Wigner negativity volume of state ρ
$P(s)$	Avalanche size distribution
$\sigma_y(\tau)$	Allan deviation at integration time τ
τ	Avalanche size exponent; integration time
$W(x, p)$	Wigner function in phase space
ξ	Stochastic perturbation vector

1 Introduction

1.1 Series architecture and naming conventions

This document is part of the Open-Source Engineering Optics Series, organized into four lettered series A–D. The *C-series* provides production-grade computational toolkits; the present document is C8. To save the reader from repeated prose lookups, Table 1 lists every series-internal identifier used in this document together with its meaning.

Table 1: Series-internal identifiers used in this document.

Code	**Family**	**Meaning**
Series A: textbook trilogy		
A1, A2, A3	Series A	Photon-lifecycle textbooks (propagate / detect / manage)
Series B: theoretical pillars		
B1	Series B	Photon stack and operator algebra
B2	Series B	Production pipeline; algorithm taxonomy
B3	Series B	Invariant audit; algorithmic invariants AI-1..15
B4	Series B	Quantum resource theory; cooperativity
B5	Series B	Chaos / extreme value statistics
B6	Series B	Path-integral parent formalism
Series C: production-grade toolkits		
C0	Series C	Shared library and architecture
C1	Series C	Hamiltonian worked examples
C2	Series C	Path-integral tolerance analysis
C3	Series C	Invariant-aware design optimization
C4	Series C	Quantum metrology and verification
C5	Series C	Computational reconstruction and inverse problems
C6	Series C	System integration for photonic circuits
C7	Series C	Benchmark suite and validation database
C8	**Series C**	**Distribution-aware production engineering (this doc)**
Series D: worked examples		
D1–D6	Series D	Worked-example database (Hamiltonian, PI, hybrid, sensors, etc.)
C8-internal codes		
M1–M6	Module	Six reusable computational engines (this doc); M6 is the meta-audit layer
TC1–TC5	Test case	Five industry-anchored demonstrations (this doc)
P1–P5	Pillar	Five production-engineering audit surfaces
T1, T2, T3	Tier	Traditional / advanced / novel implementations
I-1	Inevitability	Fundamental (quantum-statistical: Bose–Einstein)
I-2a	Inevitability	Compositional emergent (log-normal, Weibull)
I-2b	Inevitability	Self-organized emergent (power-law, $1/f$, SOC)
I-3	Inevitability	Resource (Wigner-negative quantum)
C3 algorithmic invariants used by C8 module M2		
AI-1..AI-15	Invariant	Etendue, Lagrange, diffraction limit, SBP, Petzval, unitarity, reciprocity, topological charge, etc.

Each C-series document contains *modules* (M1, M2, ...) providing reusable computational engines, and *test cases* (TC1, TC2, ...) demonstrating those modules on industry-relevant problems. C8 follows the same convention.

1.2 Why this document exists

Twentieth-century optical engineering was a coarse-grained discipline. Apertures contained many modes, detectors collected many photons, scatterers acted on many wavefronts, and fabrication errors aggregated over many surfaces. In every coarse-grained limit, the Central Limit Theorem dominates and Gaussian statistics emerge as the natural language. The legacy is RSS tolerancing, $\pm 3\sigma$ acceptance, ANOVA-based gauge R&R, Cpk capability indices, and sensitivity reports built on quadratic Taylor expansions.

Modern photonics has reversed every coarse-graining. Silicon photonic devices operate on a small number of waveguide modes; quantum photonic links use single photons; thermal photonic detectors approach the per-mode Bose–Einstein limit; co-packaged optics packs hundreds of independent channels into millimeters of substrate where a single fabrication defect can dominate yield. The CLT no longer hides the underlying physics, and the underlying physics is overwhelmingly non-Gaussian. We argue — and demonstrate quantitatively — that the distribution-aware treatment of tolerance, optimization, assembly, inspection, and capability is no longer optional in any of the three forward frontiers of photonics (silicon, thermal, quantum), and that Gaussian assumptions silently produce yield overestimates, specification mismatches, biased active-alignment convergence, and unreliable acceptance decisions.

Central Theme

Gaussianity is the exception, not the rule. It is the fixed point of coarse-graining that 20th-century optics happened to live at, and 21st-century photonics has moved past. The production-engineering toolkit of modern optical engineering must therefore be built on a *distributional ladder* of identification, fitting, robust optimization, and quantile / extreme-value specification — with Gaussian as one node on the ladder, not as the default.

1.3 Four inevitabilities of non-Gaussianity

C8 is organized around four structurally distinct mechanisms by which non-Gaussianity becomes inevitable in modern photonics (Table 2). Each mechanism has a distinct mathematical signature and a distinct engineering response.

Table 2: The four inevitabilities of non-Gaussianity in modern photonic engineering.

Code	Inevitability	Mathematical signature	Optical examples
I-1	Fundamental (quantum-statistical)	Bose–Einstein; $F = 1 + \langle n \rangle$	Thermal radiation, radiative cooling, TPV, THz radiometry
I-2a	Compositional emergent	Log-normal, Weibull (static aggregation)	Silicon photonic yield, coating reliability, mixed defect modes
I-2b	Self-organized emergent	Power-law, Pareto, $1/f^{\beta}$, SOC avalanches	Rogue waves, 1/f laser noise, SPAD after-pulsing, random lasers
I-3	Resource (quantum non-classical)	Wigner negativity; stellar rank > 0	Heralded single photons, cat states, GKP states, NOON states

The four inevitabilities have a hidden symmetry: I-1 and I-2b are "fundamental" in their respective regimes (one quantum-statistical, one classical-dynamical), bracketing two "engineering" inevitabilities I-2a and I-3 that arise respectively from aggregation operations on independent

ensembles and from quantum resources prepared by design.

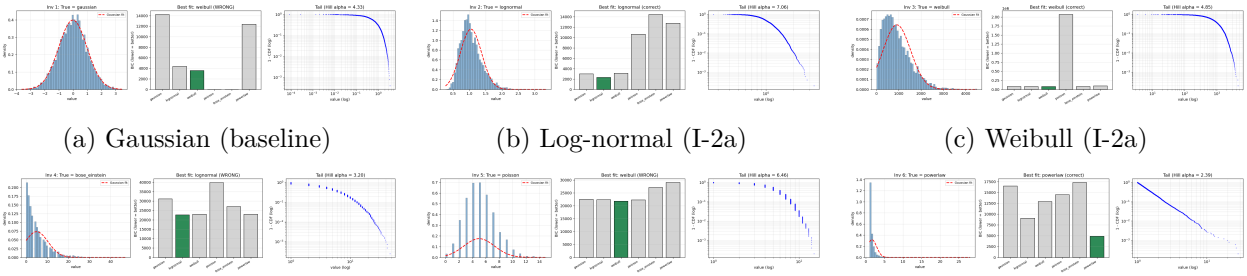
1.4 The distributional ladder: six families and where they appear

The four inevitabilities of Section 1.3 are mechanisms; what an engineer actually fits to a sample of data is one of a small number of *distribution families*. C8 module M1 (Section 9) selects among six such families using BIC with a Hill tail-index cross-check. Before meeting the test cases or the modules in detail, it is useful to put all six on one page. Table 3 lists them together with the mechanism that produces each one, the inevitability code it instantiates, and the test case in which the engineer first encounters it on the production floor.

Table 3: The six distribution families that C8 module M1 fits. "Mechanism" is the physical or compositional process that produces the family; "Inev." is the corresponding inevitability code from Table 2; "First appears in" is the test case (or module) in which the engineer first encounters the family in this document.

Family	**Signature**	**Mechanism**	**Inev.**	**First appears in**
Gaussian	$f(x) \propto e^{-x^2/2\sigma^2}$	CLT limit of additive independent errors	baseline	M2 RSS sensitivity; TC4 NM servo assumption
Log-normal	$f(x) \propto x^{-1}\, e^{-(\ln x)^2/2\sigma_{\ln}^2}$	Multiplicative aggregation: $\prod_i (1+\varepsilon_i)$ over independent factors	I-2a	TC1 MZI phase error from etch / width variation
Weibull	$f(x) \propto x^{k-1}\, e^{-(x/\lambda)^k}$	Weakest-link extreme-value distribution; aggregated failure of N stress paths	I-2a	M3 B-life reporting; coating reliability
Poisson	$P(k) = e^{-\lambda}\lambda^k/k!$	Independent rare events; shot-noise floor for low $\langle n\rangle$	baseline	TC2 (the wrong starting point for thermal NETD)
Bose–Einstein	$P(k) = (1-p)\,p^k$, $p = \langle n\rangle/(1+\langle n\rangle)$	Single-mode thermal photon statistics; bunching variance $\langle n\rangle + \langle n\rangle^2$	I-1	TC2 NETD correction by $\sqrt{F} = \sqrt{1+\langle n\rangle}$
Pareto / power-law	$f(x) \propto x^{-(\alpha+1)}$, $x \geq x_{\min}$	Self-organized / scale-free dynamics; criticality, $1/f^\beta$, SOC avalanches	I-2b	TC5(a) rogue waves; M5 SOC avalanche scaling

Figure 1 shows the shape of each of the six families on a common axis, drawn from the M1 reference run with seed 42. Reading across rows, the upper triplet collects the *continuous* families that dominate static aggregation (Gaussian, log-normal, Weibull); the lower triplet collects the *discrete* (Poisson, Bose–Einstein) and the *heavy-tailed continuous* (Pareto / power-law) families that dominate quantum-statistical and self-organized regimes. The visual contrast between the symmetric Gaussian peak and the log-tail of Pareto is the single most important pedagogical takeaway of the C8 toolkit: every C8 method exists because that contrast cannot be papered over with a wider error bar.



(a) Gaussian (baseline) (b) Log-normal (I-2a) (c) Weibull (I-2a)

(d) Poisson (baseline) (e) Bose–Einstein (I-1) (f) Pareto / power-law (I-2b)

Figure 1: The six distribution families that C8 module M1 fits, drawn from the reference simulation run (seed 42). Upper row: continuous families that dominate static aggregation regimes (Gaussian, log-normal, Weibull). Lower row: the discrete photon families (Poisson, Bose–Einstein) and the heavy-tailed Pareto / power-law family that dominates self-organized criticality. The inevitability code in each panel caption matches Table 2.

Two clarifications about Table 3 are worth stating up front. *First*, Gaussian and Poisson are listed not as "new" families but as the *baselines* that the rest of C8 corrects. They are the right answer in the coarse-grained limits where the CLT or the rare-events approximation hold (and M1 will return them when those limits do hold), and they are the wrong answer everywhere else. Their presence in the ladder is deliberate: a distribution-aware toolkit that could not detect when the Gaussian baseline is in fact correct would be just as unreliable as one that always assumed it was. *Second*, the four inevitabilities of Section 1.3 have only *three* representatives in this table (I-1, I-2a, and I-2b); I-3, the resource inevitability, is intentionally absent. The reason is that Wigner negativity is a property of a 2D quasi-probability $W(x, p)$ on phase space, not of a 1D sample histogram, and therefore cannot be fit by any of the six 1D estimators that M1 ranks. I-3 is handled by a separate channel in C8: the Wigner-tomography station of M5 (Section 13, T3) and the operating-map analysis of TC3 (Section 6). The reader should think of the six-family ladder of Table 3 as the *classical* distributional ladder, with I-3 sitting on a parallel quantum track.

Figure 18, presented later in the M1 section, shows where the six families separate in the skewness–excess-kurtosis plane and is the visual companion to Table 3. The reader who wants the picture now can refer forward; the rest of this introduction needs only the verbal mapping the table provides.

1.5 Five pillars of production-engineering audit

C8 audits five pillars of any production-engineering program, each of which encodes an implicit Gaussian assumption that must be revised. Table 4 lists the pillars together with the hidden assumption and the distribution-aware replacement.

Table 4: Five-pillar audit surface of distribution-aware production engineering.

Pillar	Hidden Gaussian assumption	Distribution-aware replacement
P1 Design baseline	Nominal peak = best operating point	Robust-optimal point under CVaR or trimmed mean (M2)
P2 Specification	$\pm n\sigma$ windows	Quantile / CVaR / B-life / Wigner-negativity language (M3)
P3 Assembly machines	Quadratic cost surface, deterministic machine	Bayesian active alignment + two-distribution deconvolution (M4)
P4 Inspection	Uniform sampling, ANOVA gauge R&R	Tail-aware sampling, non-parametric R&R, quantum tomography (M5)
P5 Capability reporting	Cpk, PPM, 6σ	Quantile-Cpk, B-life, Fano factor, negativity volume (M3)

1.6 Production-engineering workflow with non-Gaussianity entry points

Figure 2 synthesises the four inevitabilities, the five pillars, and the five C8 modules into a single production pipeline. Reading left to right: a product flows from *design baseline* through *specification*, *assembly*, *inspection*, and *capability reporting*, eventually into the field. Each pillar receives non-Gaussian inputs (top badges) and is served by one or more C8 modules (bottom boxes). The dashed feedback arc represents field-return data flowing back into design baseline updates.

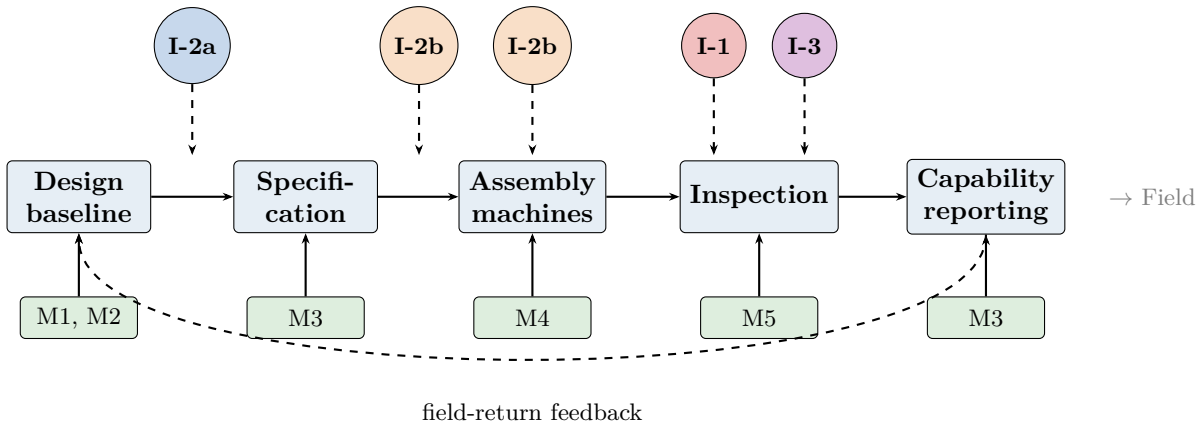


Figure 2: C8 production-engineering workflow. Pillars P1–P5 are shown horizontally (blue boxes); C8 modules M1–M5 attach below; the four inevitabilities I-1, I-2a, I-2b, I-3 enter as colored badges at their respective pillars. Color coding matches Table 2.

1.7 Document structure

Section 2 sets the C8 scope boundary and the C3-to-C8 stochastic duality. Section 3 defines the three-tier (T1 / T2 / T3) architecture shared with the rest of the C-series. Sections 4–8 present the five test cases, *placed before the modules* following the series pedagogical principle that practitioners encounter their engineering problem before the mathematical machinery. Sections 9–14 present the six modules, of which M6 is a meta-layer that audits the statistical quality of the risk metrics produced by M1–M5. Section 15 provides the cross-test-case comparison. Section 17 covers what C8 does *not* cover. Section 18 documents reproducibility. The bibliography follows.

2 Scope Boundary

2.1 What C8 covers

C8 implements the *production-engineering layer* on top of existing forward-design (C3), tolerance (C2), metrology (C4), reconstruction (C5), system-integration (C6), and benchmark (C7) toolkits. Its scope is intentionally cross-cutting: distribution- aware methods touch every stage of the B2 pipeline, from design freeze through specification, assembly, inspection, and acceptance. C8 does not replace any of the existing C-series toolkits; it sits above them and augments their statistical layer.

2.2 C3–C8 stochastic duality

C5 was introduced as the inverse-problem dual of C3's forward problem: where C3 designs subject to invariants, C5 reconstructs subject to those same invariants used as priors. C8 introduces a *stochastic* dual to C3, organized along a different axis (Table 5).

Table 5: C3 deterministic forward design vs C8 stochastic production engineering.

	C3 (deterministic / forward)	**C8 (stochastic / production)**
Object	Nominal design x_0	Distribution $P(\xi)$ around x_0
Math	Invariant catalogue, feasibility gates	Characteristic / Wigner function, moment + tail tests
Merit	$\min f(x_0)$ subject to invariants	$\min \mathrm{CVaR}_\alpha[f(x_0+\xi)]$ or $\min \mathrm{TM}_\alpha[\cdot]$
Output	Optimized design	Robust-optimal + specs + inspection plan
Metric	Strehl, MTF, fidelity	Quantile-Cpk, B-life, $\mathcal{N}(\rho)$

The same invariant catalogue used by C3 as feasibility gates is re-used by C8 module M2 as feasibility projections inside the robust-optimization layer, so that distribution-aware optima remain physically realizable.

2.3 What C8 does not cover

Honest Assessment

Current version of C8 covers *methodology and reference implementations* on synthetic data. It does not cover the items in Table 6, which are deferred to future versions or to other documents in the series.

Table 6: Items deliberately out of scope for C8's current version.

Excluded item	**Where it belongs**	**Reason**
Foundry-specific PDK statistics	Foundry-specific	Proprietary process data
Experimental field validation	Future version of C8	Requires partner data
Full DRO mathematical theory	B-series future	Out of toolkit scope
Contractual / legal language revision	Industry white paper	Non-technical
Non-photonic reliability (MEMS, ICs)	Out of scope	Photonic-specific
Full quantum error correction	C4 v3	Scope overreach

3 Three-Tier Architecture

C8 follows the three-tier architecture established in C3, C4, and C5. Each module provides three nested implementations of increasing capability and decreasing familiarity to traditional practice (Table 7).

Table 7: Three-tier master table for C8 modules.

	T1 Traditional	T2 Advanced	T3 Novel
M1	Gaussian-only fit	Multi-family BIC selection	Pareto + Hill cross-check, Wigner / characteristic function
M2	RSS sensitivity	Monte Carlo CVaR + DRO	Trim-mean fallback under power-law, invariant projection
M3	Cpk	Box-Cox / quantile-Cpk	Fano, B-life, $1/f$ corner, negativity-based spec
M4	Quadratic hill-climb (Nelder–Mead)	Bayesian optimization	Joint part / machine deconvolution
M5	Uniform sampling, ANOVA R&R	Importance + SPRT, quantile R&R	SOC avalanche scaling + Wigner tomography station
M6	Single point estimate of any risk number	Nonparametric bootstrap CI; Hill threshold tuning; parametric-bootstrap B-life correction	Mis-specification regret matrix in USD/year; sample-size planning vs tail index

4 Test Case 1: Silicon Photonic MZI Network Yield (I-2a)

The first test case anchors C8 in the largest single industrial deployment of silicon photonics today: co-packaged optics (CPO) for AI / datacenter interconnects at 800G and 1.6T. The MZI network is the workhorse of every CPO transceiver. Its yield sensitivity to fabrication phase error is well known to foundry engineers but is almost universally analyzed under a hidden Gaussian assumption that systematically overstates yield. TC1 shows that the same data, treated with distribution-aware methods, gives a much more honest forecast and unlocks a pre-bias trim that recovers a sizeable fraction of the silently lost margin.

TC1 Scope

Device: 16-channel SOI Mach–Zehnder interferometer network for 800G co-packaged optics, $\lambda = 1310\,\mathrm{nm}$, ER target $= 20\,\mathrm{dB}$.
Industry pain point: Width and etch variations propagate multiplicatively through $\delta\phi = \beta\,\delta L + \delta\beta\,L$, producing log-normal phase errors. Gaussian Cpk over-estimates yield, leading to silent yield loss at the foundry interface.
Pillars exercised: P1 (pre-bias), P2 (quantile spec), P5 (quantile-Cpk).
Modules exercised: M1, M2, M3.
Inevitability: I-2a (compositional emergent).

4.1 System layout

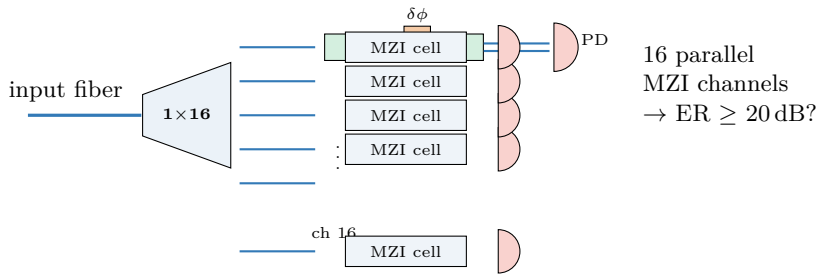


Figure 3: TC1 system layout. A single input fiber feeds a 1×16 splitter; each output drives an MZI cell consisting of two 2×2 couplers (green) and a heater (orange) on one arm. The phase imbalance $\delta\phi$ from the ideal π setpoint determines the per-channel extinction ratio at the output photodetectors.

4.2 Model and metrics

The MZI extinction ratio depends on the phase imbalance $\delta\phi$ from the ideal π setpoint:

$$\mathrm{ER}(\delta\phi) = 20\,\log_{10}\left|\frac{\cos(\delta\phi/2)}{\sin(\delta\phi/2)}\right| \quad [\mathrm{dB}]. \tag{1}$$

The phase error is modelled as

$$\delta\phi = \mu_b + \mathrm{sign}(s)\,\eta\cdot\mathcal{L}(\sigma_{\ln}), \tag{2}$$

where $\mathcal{L}(\sigma_{\ln})$ denotes a unit-mean log-normal random variable with log-space standard deviation $\sigma_{\ln}$, η is a per-channel scale, s is a random sign, and μ_b is an optional pre-bias offset. Channel yield is the fraction of channels with ER $\geq$ 20 dB. System yield is the fraction of devices in which all 16 channels pass.

4.3 Results: log-normal vs Gaussian-Cpk prediction

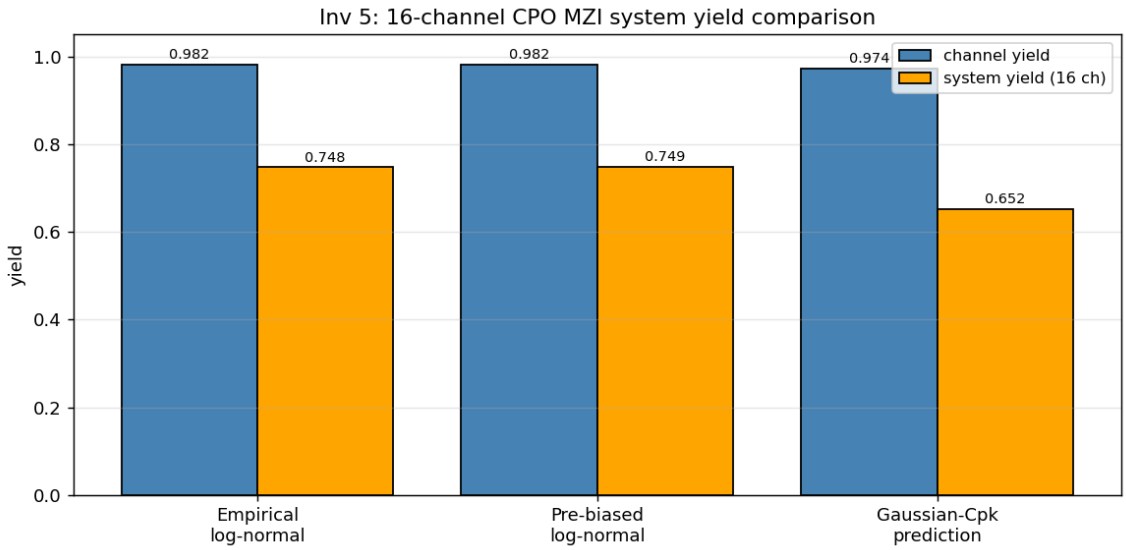


Figure 4: TC1 Investigation 5: 16-channel CPO MZI system yield. Empirical log-normal yield, pre-biased log-normal yield, and the yield that a Gaussian-Cpk calculation would predict using the same observed phase standard deviation. The Gaussian-Cpk prediction *systematically over-states* both channel and system yield.

4.4 Pre-biasing under log-normal phase error

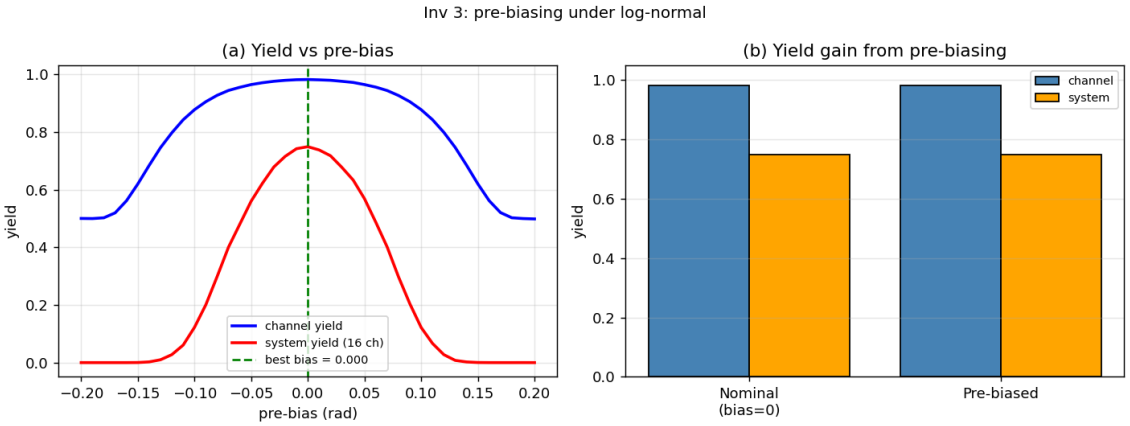


Figure 5: TC1 Investigation 3: yield versus pre-bias. The asymmetric log-normal phase error pushes the yield-optimum *away* from the nominal π setpoint.

Design Rule

TC1 design rule. For any optical phase setting whose fabrication error is log-normal, the production-optimal phase *is not* the nominal phase. Compute the pre-bias offset $\Delta x^* = \arg\min \mathrm{CVaR}_\alpha[f(x_0+\xi)] - \arg\max f(x_0)$ from M2 and apply it as a pre-trim during device tuning.

4.5 TC1 figures of merit

Table 8: TC1 figures of merit.

FoM	Order of magnitude	Why
Channel yield	0.7–0.999	Per-MZI acceptance fraction
System yield (16 ch)	0.001–0.95	Multiplicative penalty of channel yield
Pre-bias Δx^*	10^{-2}–10^{-1} rad	Asymmetry-driven offset
Cpk overestimate	1–20 percentage points	Failure of Gaussian assumption

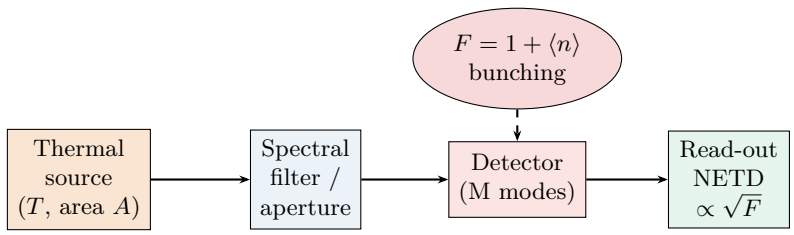
5 Test Case 2: Thermal-Photonic NETD with Bose–Einstein (I-1)

The second test case enters the regime where non-Gaussianity is not engineering pragmatics but a law of nature. Thermal photons obey Bose–Einstein statistics with bunching variance $\langle n\rangle^2$, and any radiometric system that operates at non-negligible mean occupation per mode — which includes mid-IR atmospheric-window radiometry, TPV, and any room- temperature far-IR / THz application — has a Fano factor greater than one and a noise budget that the standard Poisson NETD formula understates. TC2 quantifies the correction across three industrially relevant regimes.

TC2 Scope

Device: (a) Daytime radiative cooler, 8–13 μm, $T_{\text{scene}} = 300\,\text{K}$; (b) TPV emitter at $T = 1800\,\text{K}$, $\lambda = 2\,\mu\text{m}$; (c) THz radiometer at $\lambda = 300\,\mu\text{m}$, $T = 300\,\text{K}$.
Industry pain point: Standard NETD formulas assume Poisson statistics. The Bose–Einstein photon-number variance $\mathrm{Var}(n) = \langle n\rangle + \langle n\rangle^2$ adds a *bunching term* that dominates whenever $\langle n\rangle$ per mode is appreciable, but is silently absorbed into "noise margin."
Pillars exercised: P2 (Fano-corrected spec), P5 (Fano reporting).
Modules exercised: M1, M3.
Inevitability: I-1 (fundamental).

5.1 System layout



(a) cooler, λ=10μm
(b) TPV, λ=2μm
(c) THz, λ=300μm

Figure 6: TC2 system layout (shared abstraction). A thermal source of temperature T illuminates a spectral filter / aperture; the transmitted photons reach a detector with M collected modes. The Bose–Einstein bunching term $F = 1 + \langle n \rangle$ enters at the detector and inflates the read-out NETD by $\sqrt{F}$ relative to a Poisson estimate. The same diagram instantiates three regimes (a)–(c).

5.2 Bose–Einstein photon statistics

For a single mode in thermal equilibrium at temperature T and optical frequency ν,

$$\langle n \rangle = \frac{1}{e^{h\nu/k_B T} - 1}, \qquad \mathrm{Var}(n) = \langle n \rangle + \langle n \rangle^2, \qquad F \;=\; \frac{\mathrm{Var}(n)}{\langle n \rangle} \;=\; 1 + \langle n \rangle. \tag{3}$$

5.3 Results: Fano spectrum and THz regime

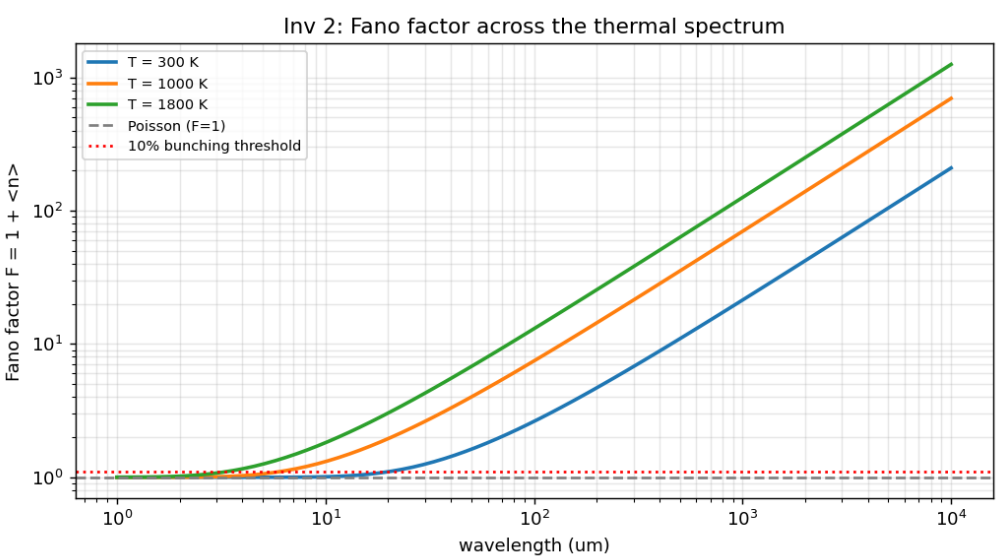


Figure 7: TC2 Investigation 2: Fano factor $F = 1 + \langle n \rangle$ across the thermal IR / THz spectrum at three temperatures.

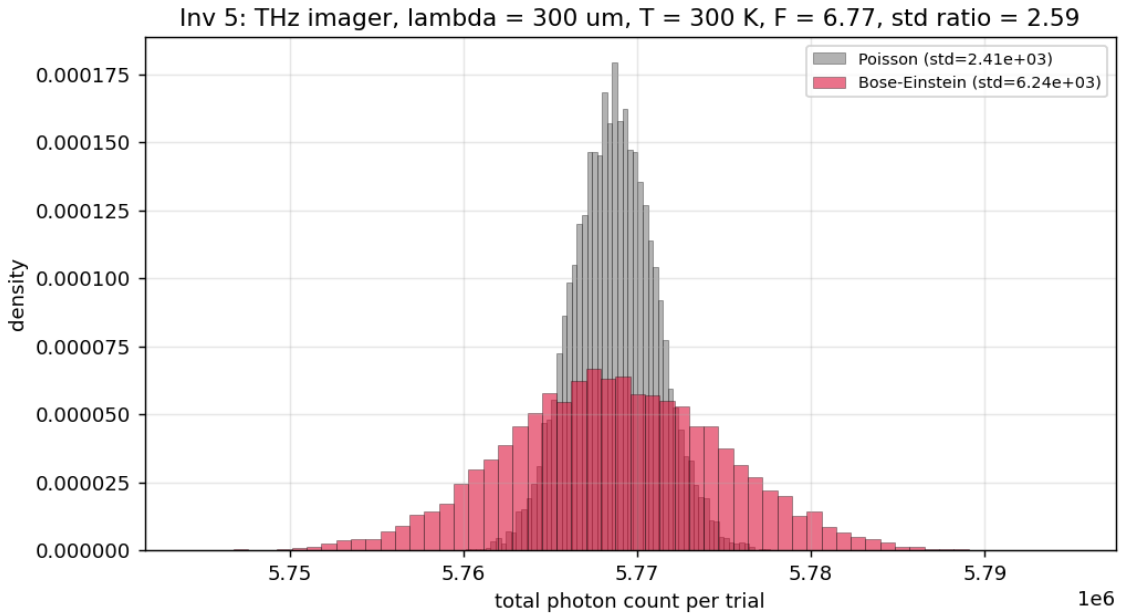


Figure 8: TC2 Investigation 5: photon-count distributions for a THz radiometer ($\lambda = 300\,\mu$m, $T = 300\,$K).

Central Finding

For any thermal-photonic product whose mean photon occupation per mode exceeds $\langle n \rangle \approx 0.1$, NETD must be reported with the Fano correction $F = 1 + \langle n \rangle$. This is non-negotiable for THz / sub-mm radiometry and important even in the mid-IR atmospheric window for high-precision radiative coolers.

6 Test Case 3: Heralded Single-Photon Wigner Negativity (I-3)

The third test case targets the third industrial frontier of modern photonics: quantum sources for QKD and photonic quantum computing. Practitioners routinely certify "single-photon" sources with the antibunching test $g^{(2)}(0) < 1$, but $g^{(2)}(0)$ is a second-order intensity correlation and is not sensitive to the quasi-probability features that distinguish a genuinely non-classical state from a Gaussian / squeezed substitute. TC3 shows that the right specification quantity is the Wigner negativity volume $\mathcal{N}(\rho)$, that $\mathcal{N}$ and $g^{(2)}(0)$ are not equivalent, and that an in-line Wigner tomography station can serve as a production- floor inspection tool.

TC3 Scope

Device: Heralded single-photon source from SPDC in PPLN.
Industry pain point: "Single-photon" sources are classically certified by $g^{(2)}(0) < 1$, but $g^{(2)}(0)$ does *not* certify the genuinely non-classical resource needed for quantum advantage.
Pillars exercised: P2 (negativity spec), P4 (quantum inspection), P5 (stellar-rank reporting).
Modules exercised: M1, M3, M5.
Inevitability: I-3 (resource).

6.1 System layout

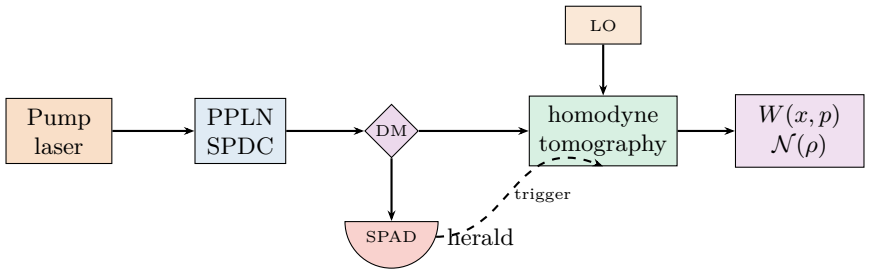


Figure 9: TC3 system layout. Pump laser drives spontaneous parametric down-conversion in a PPLN waveguide. A dichroic mirror separates signal and idler. Detection of the idler in a SPAD ("herald") triggers Wigner tomography on the signal via a homodyne measurement against a local oscillator (LO). The output is the reconstructed Wigner function $W(x,p)$ and its negativity volume $\mathcal{N}(\rho)$.

6.2 Wigner functions of Fock-diagonal states

For a Fock-diagonal mixed state $\rho = \sum_n p_n |n\rangle\langle n|$, the Wigner function is

$$W(x,p) \;=\; \frac{1}{\pi}\sum_n p_n\,(-1)^n\,L_n(2(x^2+p^2))\,e^{-(x^2+p^2)}, \tag{4}$$

where L_n is the Laguerre polynomial of order n. The *Wigner negativity volume* of the state is

$$\mathcal{N}(\rho) \;=\; \tfrac{1}{2}\left(\int |W(x,p)|\,dx\,dp \;-\; 1\right), \tag{5}$$

which is zero for any classical / Gaussian state and strictly positive for any genuinely non-classical state.

6.3 Loss and multi-pair operating map

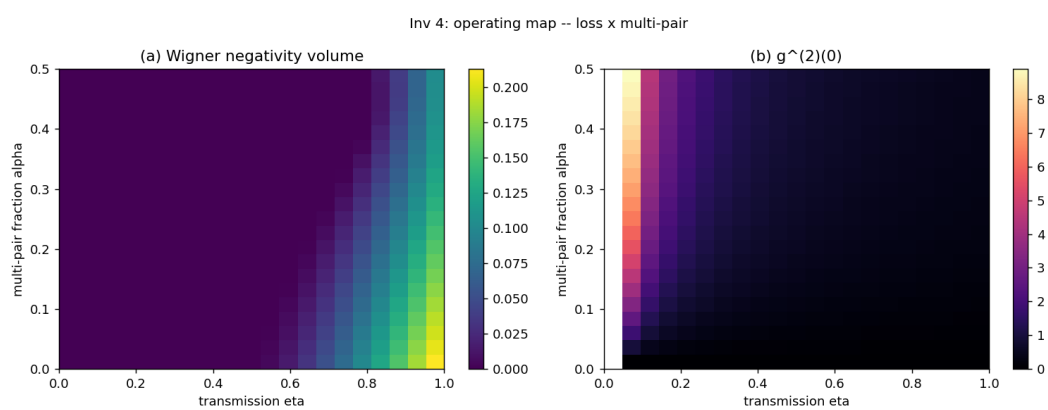


Figure 10: TC3 Investigation 4: 2D operating map of Wigner negativity volume (left) and $g^{(2)}(0)$ (right) versus heralding transmission η and multi-pair contamination α.

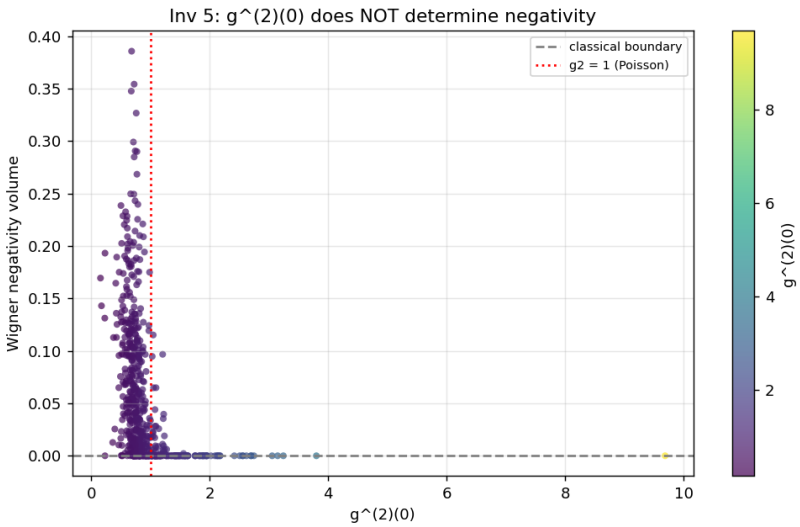


Figure 11: TC3 Investigation 5: 600 random Fock-diagonal mixtures in the $g^{(2)}(0)$–$\mathcal{N}(\rho)$ plane. The two metrics are not equivalent.

Design Rule

TC3 design rule. The acceptance criterion for any quantum-photonic source intended for QKD or photonic quantum computing must include a positive Wigner negativity volume, not merely $g^{(2)}(0) < 1$. The negativity volume should be reported as a number on the datasheet and certified by an in-line Wigner tomography station (M5 T3).

7 Test Case 4: Active Fiber-to-PIC Alignment (Cross-cutting)

The fourth test case is the production-floor anchor of the document. Every CPO module shipped today goes through an active fiber-to-PIC alignment step on a stage controlled by a cost- function servo. The default servo — Nelder–Mead or other quadratic-cost methods — assumes both that the optical landscape is smooth and convex *and* that the measurement noise of the alignment machine is Gaussian. Both assumptions fail in real fabs, where mode mismatch generates asymmetric landscapes and thermal cycling / air-bearing events generate bimodal or heavy-tailed measurement traces. TC4 shows the biased convergence of the standard servo and demonstrates that a minimal Bayesian-optimization replacement recovers near- Gaussian alignment quality.

TC4 Scope

Device: Fiber array to 16-channel SOI waveguide array in CPO packaging step.
Industry pain point: Active-alignment machines use quadratic hill-climbing servos. When the coupling landscape is asymmetric (mode-mismatch foundry variation) and the machine noise is non-Gaussian, the servo converges to biased points and time-to-alignment grows.
Pillars exercised: P3 (Bayesian alignment), P4 (machine / part deconvolution).
Modules exercised: M1, M4, M5.
Inevitability: cross-cutting (touches I-2a and I-2b).

7.1 System layout

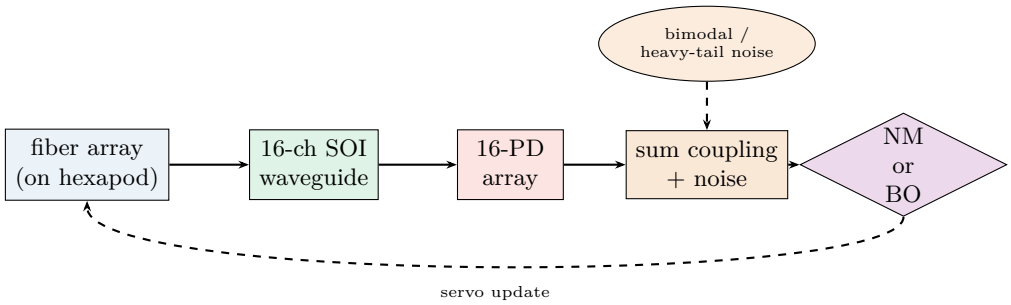


Figure 12: TC4 system layout. The fiber array on a hexapod stage is aligned to a 16-channel SOI PIC. The summed coupling readout plus machine noise drives a controller that is either Nelder–Mead (default) or Bayesian optimization (M4 T2). Bimodal or heavy-tailed machine noise enters at the readout.

7.2 Convergence bias under bimodal and heavy-tailed noise

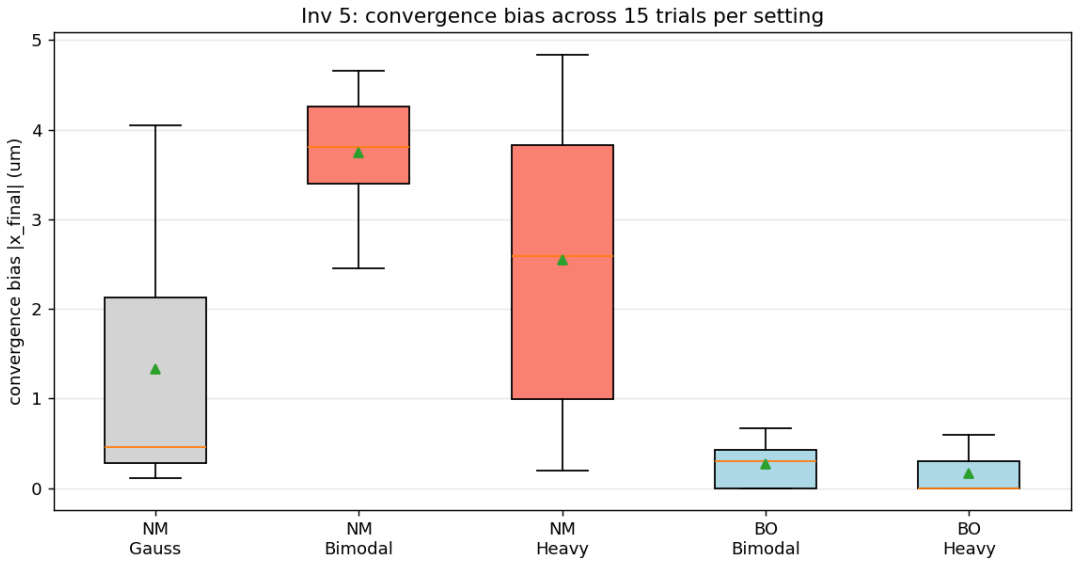


Figure 13: TC4 Investigation 5: 15-trial robustness comparison. Nelder–Mead vs Bayesian optimization across Gaussian, bimodal, and Student-t noise models.

Central Finding

Active-alignment machines that operate under non-Gaussian machine-noise distributions should replace Nelder–Mead with Bayesian optimization with Gaussian-process surrogate. The GP-LCB acquisition averages out local jitter via posterior smoothing and recovers near-Gaussian convergence quality on bimodal and heavy-tailed measurement channels.

8 Test Case 5: Self-Organized Photonics (I-2b)

The fifth test case closes the four-inevitability loop by addressing the family of phenomena that none of the previous four test cases captures: scale-free, dynamical, power-law behavior arising from feedback and criticality rather than from aggregation or quantum statistics. Two photonic instances are demonstrated — optical rogue waves in supercontinuum and 1/f frequency noise in semiconductor lasers — and both are shown to break standard Gaussian / log-normal specification language. The right replacements are Hill tail index for the rogue intensity distribution and Allan deviation for the laser frequency noise.

TC5 Scope

Devices: (a) Noise-seeded supercontinuum proxy (rogue waves); (b) semiconductor laser frequency-noise time series ($1/f^{\beta}$ + white floor).
Industry pain point: Both phenomena are scale-free / power-law / dynamical, and neither is captured by Gaussian, log-normal, or Weibull models.
Pillars exercised: P2 (quantile / Allan-deviation spec), P4 (peak-event sampling), P5 (corner-frequency report).
Modules exercised: M1, M3, M5.
Inevitability: I-2b (self-organized).

8.1 System layout

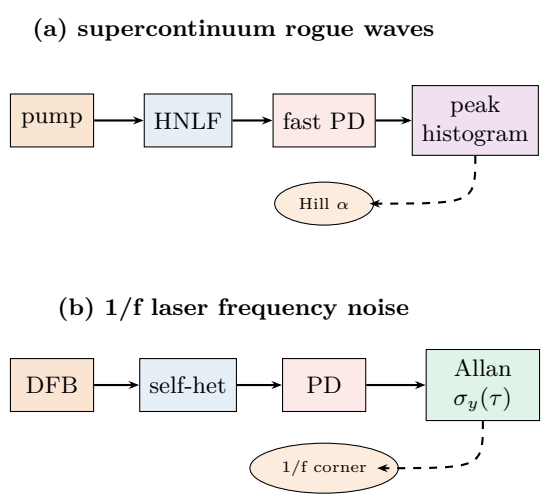


Figure 14: TC5 system layout. (a) Pump laser pumps a highly nonlinear fiber (HNLF); the output is read by a fast photodiode and a peak-intensity histogram is built. The Hill tail index α on the histogram tail is the rogue-wave diagnostic. (b) DFB laser is fed into a self-heterodyne MZI; the photodiode output goes into an Allan deviation analyzer. The 1/f corner frequency is the diagnostic.

8.2 Optical rogue waves: heavy-tailed peak statistics

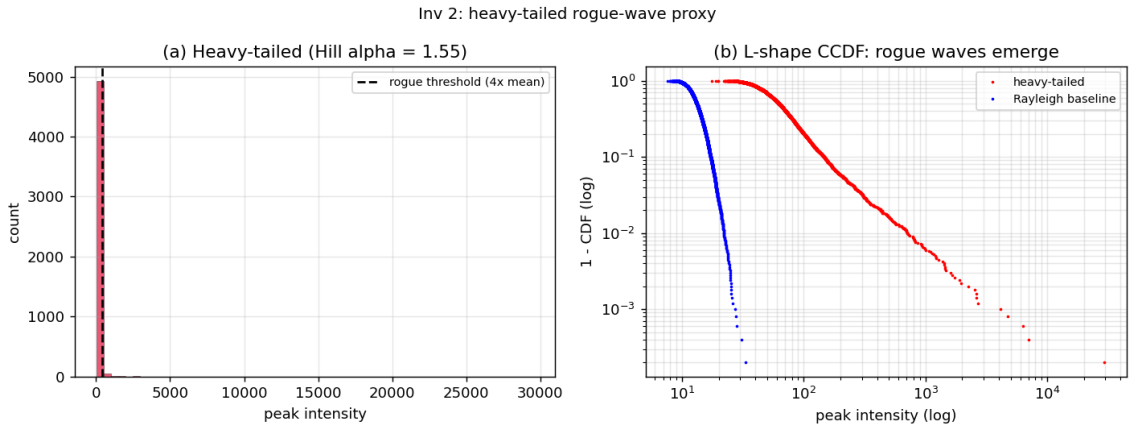


Figure 15: TC5 Investigation 2: peak-intensity statistics for the rogue-wave proxy.

8.3 1/f laser frequency noise: PSD and Allan deviation

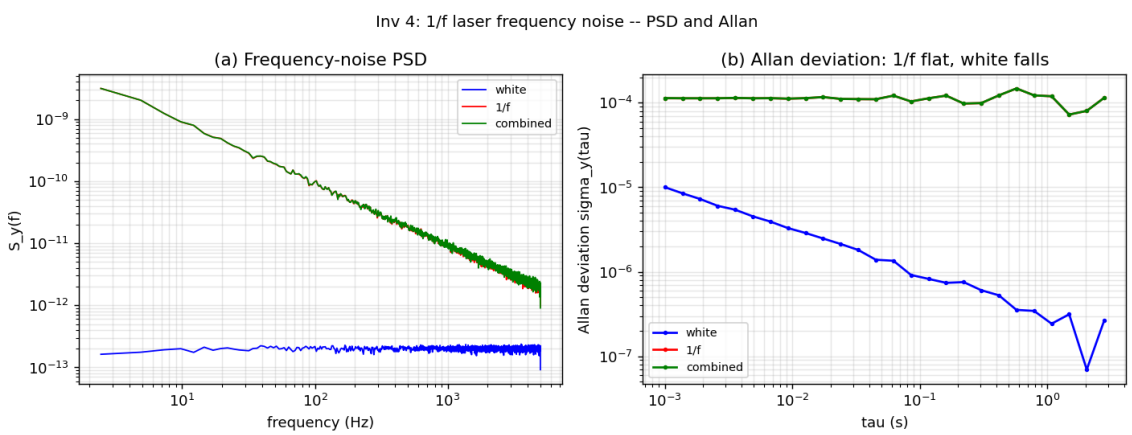


Figure 16: TC5 Investigation 4: PSD and Allan deviation of laser frequency noise.

Design Rule

TC5 design rule. For laser frequency noise, specify the Allan deviation $\sigma_y(\tau)$ as a curve on log–log axes, together with the corner frequency where the $1/f^\beta$ tail meets the white-noise floor. Do *not* report linewidth as a single Gaussian standard deviation.

9 Module M1: Distribution Identification & Classification

M1 Scope

Take any sample of size N (Monte Carlo output, in-line inspection log, instrument trace) and decide which distribution family generated it. Report moment and tail diagnostics, fit a ladder of six candidate families (Gaussian, log-normal, Weibull, Poisson, Bose–Einstein / geometric, Pareto / power-law), and select the best by BIC. Cross-check the BIC winner against the Hill tail-index estimator. Issue a CVaR-validity warning when $\hat{\alpha}_{\text{Hill}} \leq 2$.

9.1 Workflow

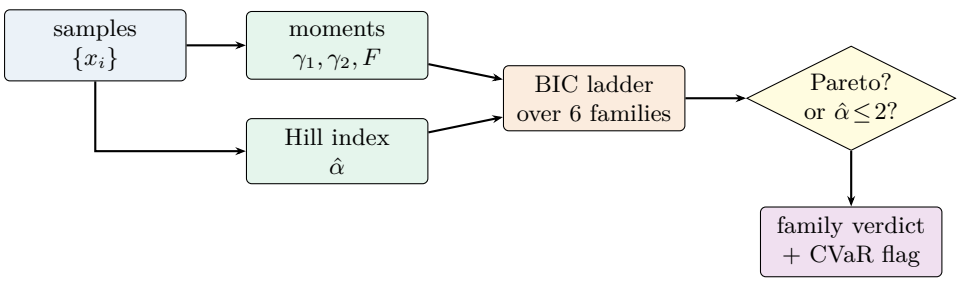


Figure 17: M1 workflow: samples in, moments and Hill index computed in parallel, BIC ladder over six families, cross-check gate, family verdict and CVaR-validity flag out.

9.2 Algorithm

For sample $\{x_i\}_{i=1}^{N}$, M1 computes:

1. Moment diagnostics: mean, std, skewness γ_1, excess kurtosis γ_2, Fano factor.
2. Gaussianity tests: KS and Anderson–Darling against the fitted normal; KL divergence to the fitted normal.
3. Hill tail index $\hat{\alpha}_{\text{Hill}}$ on the upper decile of positive samples.
4. Maximum-likelihood fit and BIC score for each candidate family in {Gaussian, log-normal, Weibull, Poisson, Bose–Einstein, Pareto}.
5. Selection of the lowest-BIC family, with cross-check against $\hat{\alpha}_{\text{Hill}}$ when the winner is Pareto or log-normal.

9.3 Validation

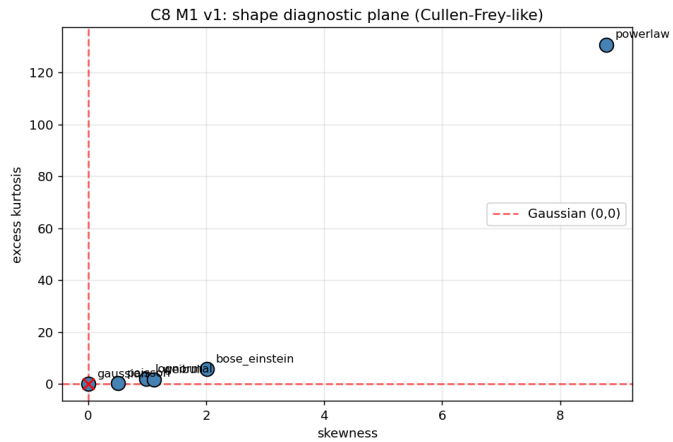


Figure 18: M1 shape diagnostic plane (Cullen–Frey-like): the six ground-truth families occupy distinct regions of the skewness–excess-kurtosis plane.

10 Module M2: Robust / Distributionally Robust Optimization

M2 Scope

Replace the deterministic merit function $f(x)$ of C3 with a robust merit functional that operates on the distribution $P(\xi)$ supplied by M1. Provide three nested merit options: expected loss (T1), CVaR$_\alpha$ (T2), and Wasserstein-DRO worst case (T3). When the perturbation distribution has Hill index ≤ 2, fall back to a symmetric trimmed mean.

10.1 Workflow

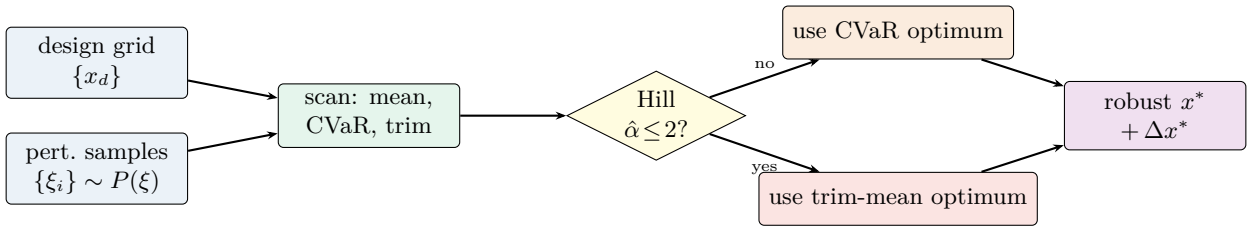


Figure 19: M2 workflow: design grid and perturbation samples enter in parallel; the scanner computes mean, CVaR and trim-mean at each design point; a Hill-index gate decides between the CVaR optimum and the trim-mean fallback; the output is the robust design point and its pre-bias offset.

10.2 Risk metrics

For a perturbation distribution $P(\xi)$ inducing a loss distribution $L(\xi) = f(x+\xi) - f(x)$,

$$\mathrm{CVaR}_\alpha[L] = \mathbb{E}[L \mid L \geq \mathrm{VaR}_\alpha[L]], \tag{6}$$

$$\mathrm{TM}_\alpha[L] = \frac{1}{N(1-2\alpha)} \sum_{i=k+1}^{N-k} L_{(i)}, \tag{7}$$

where $L_{(i)}$ is the i-th order statistic and $k = \lfloor \alpha N \rfloor$.

Bridge Equation

CVaR validity bridge. For any random variable X with power-law tail $P(X > x) \sim x^{-\alpha}$ and $\alpha \leq 2$, the sample CVaR estimator has unbounded variance. The trimmed-mean estimator TM_α is a consistent finite-variance estimator of a comparable risk quantity in the same regime, and should be substituted automatically when M1 reports $\hat{\alpha}_{\mathrm{Hill}} \leq 2$.

10.3 Pre-bias offset under power-law perturbation

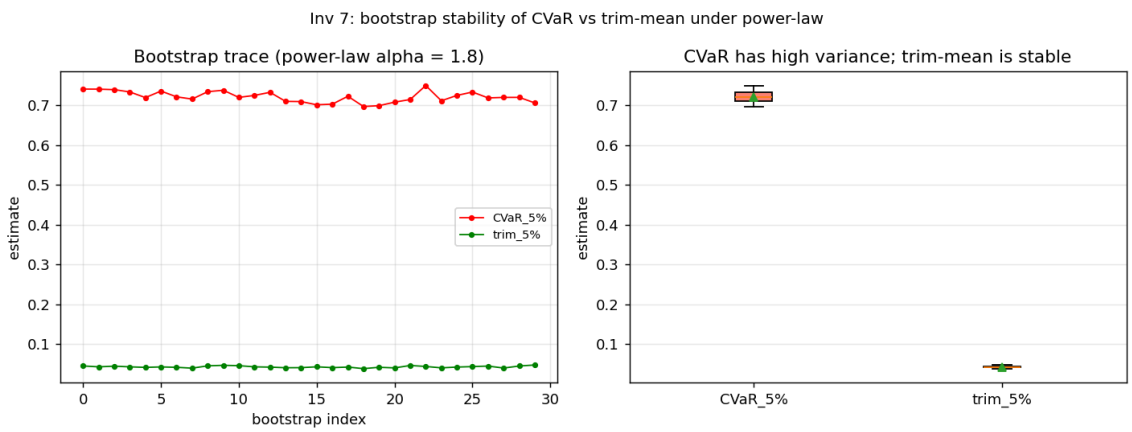


Figure 20: M2 Investigation 7: bootstrap stability of CVaR vs trimmed mean under power-law perturbation.

11 Module M3: Non-Gaussian Capability & Specification

M3 Scope

Replace the Gaussian process capability index Cpk by three distribution-aware capability indices: quantile-Cpk (Cpk_q), Box–Cox transformed Cpk, and direct empirical yield. Add distribution-native specification quantities: B-life (Weibull), Fano factor (Bose–Einstein), Hill-α (Pareto), and 1/f corner frequency (PSD-fitted).

11.1 Workflow

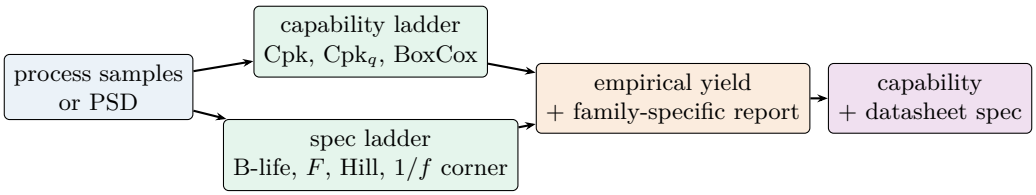


Figure 21: M3 workflow: process samples (or PSD) feed both the capability ladder and the spec ladder in parallel; the merged report combines empirical yield with the family-specific specification.

11.2 Capability indices

The standard Cpk index

$$\mathrm{Cpk} \;=\; \min\left(\frac{\mathrm{USL}-\mu}{3\sigma},\; \frac{\mu-\mathrm{LSL}}{3\sigma}\right) \tag{8}$$

is well-defined only for processes with finite mean and finite variance and is interpretable only for processes that are approximately Gaussian. The quantile-based replacement

$$\mathrm{Cpk}_q \;=\; \min\left(\frac{\mathrm{USL}-Q_{0.5}}{Q_{0.99865}-Q_{0.5}},\; \frac{Q_{0.5}-\mathrm{LSL}}{Q_{0.5}-Q_{0.00135}}\right) \tag{9}$$

reduces to (8) for Gaussian data and remains distribution-agnostic in general.

11.3 Cpk overestimate of yield on non-Gaussian processes

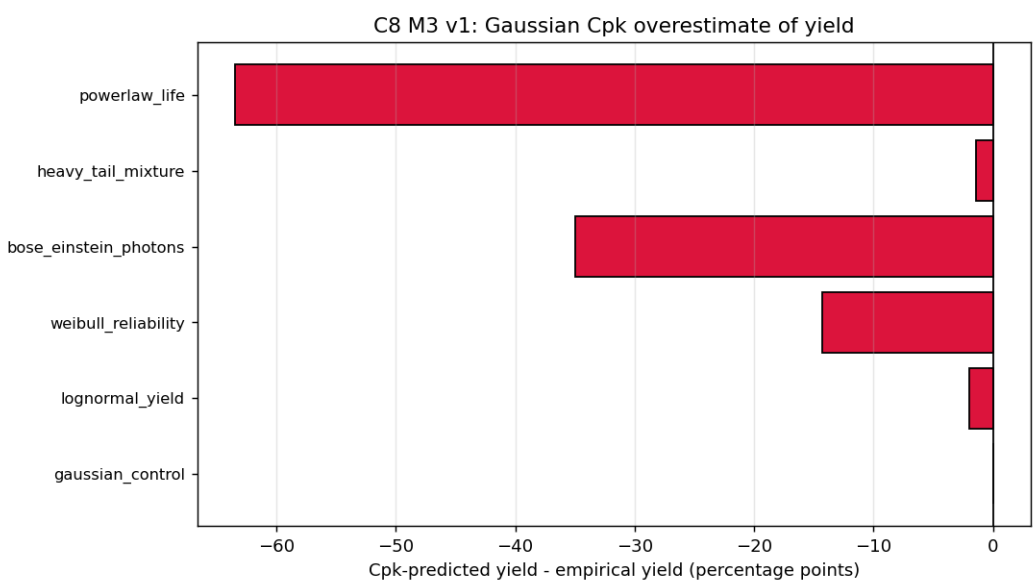


Figure 22: M3 yield-gap chart across seven investigations.

12 Module M4: Distribution-Aware Assembly Control

M4 Scope

Provide two services for active-alignment machines: (i) a Bayesian-optimization servo loop that replaces Nelder–Mead quadratic hill-climbing under non-Gaussian cost surfaces or non-Gaussian measurement noise; (ii) a two-distribution deconvolution routine that separates part variability from machine noise via parametric joint maximum likelihood.

12.1 Workflow

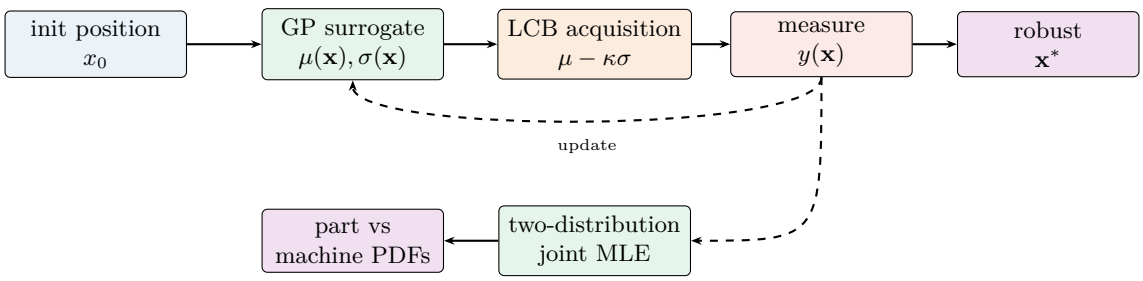


Figure 23: M4 workflow. Top branch: Bayesian-optimization servo loop with GP surrogate, LCB acquisition, measurement, and posterior update. Bottom branch: two-distribution joint MLE on the measurement trace separates part variability from machine noise.

12.2 Bayesian active alignment

The minimal Gaussian-process surrogate uses an RBF kernel with length scale ℓ and signal scale σ_f, with

$$K_{ij} \;=\; \sigma_f^2 \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\ell^2}\right) + \sigma_n^2 \delta_{ij}, \tag{10}$$

and acquisition by lower-confidence-bound, $\mathrm{LCB}(\mathbf{x}) = \mu(\mathbf{x}) - \kappa\sigma(\mathbf{x})$.

12.3 Two-distribution deconvolution

For an observed scalar $y = p + m$, where p is the part variability and m is the machine noise, the joint maximum-likelihood estimator finds parameters $\boldsymbol{\theta} = (\theta_p, \theta_m)$ minimizing

$$\boldsymbol{\theta}^* \;=\; \arg\min_{\boldsymbol{\theta}} -\sum_i \log\left(p_{\theta_p} * q_{\theta_m}\right)(y_i), \tag{11}$$

where $*$ denotes convolution.

13 Module M5: Tail-Aware Inspection & Gauge

M5 Scope

Five capabilities: (i) importance-sampling tail-probability estimator; (ii) Wald sequential probability ratio test for lot acceptance; (iii) non-parametric quantile gauge R&R for log-normal repeated measurements; (iv) Wigner-tomography inspection-station reference design; (v) self-organized criticality (SOC) avalanche statistics with finite-size scaling collapse.

13.1 Workflow

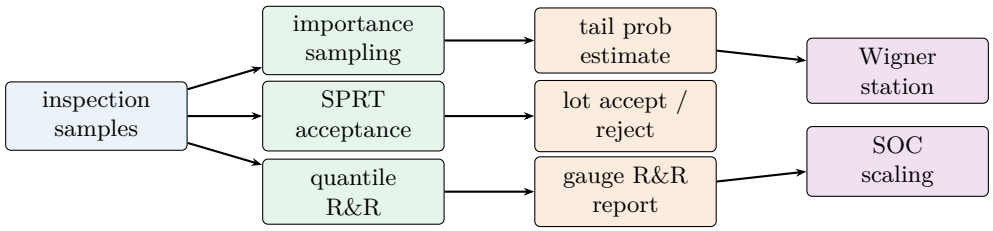


Figure 24: M5 workflow. Inspection samples branch into three parallel diagnostics (importance sampling, SPRT, quantile R&R), each producing a per-pillar report. Two T3 capabilities (Wigner tomography station and SOC avalanche scaling) sit on top of the standard diagnostics.

13.2 Wigner-tomography inspection station

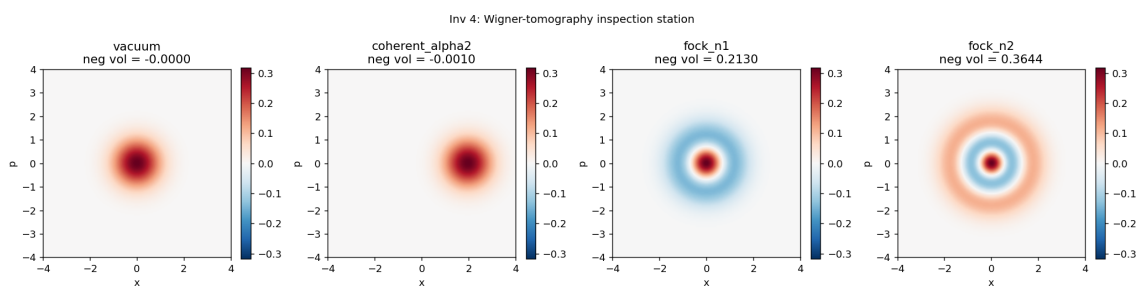


Figure 25: M5 Investigation 4: Wigner functions for the four reference states of the inspection station.

13.3 SOC avalanche scaling

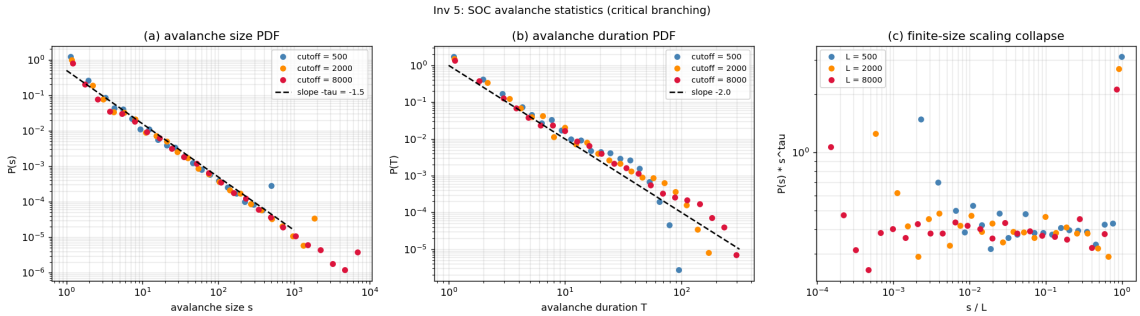


Figure 26: M5 Investigation 5: avalanche statistics from a critical Galton–Watson branching process at three cutoff sizes, with finite-size scaling collapse.

14 Module M6: Risk Quality and Mis-specification Cost

M6 Scope

M6 is the meta-layer of C8: it audits the *statistical quality* of every risk metric produced by modules M1–M5 and translates the cost of choosing the wrong distribution family into yield loss and USD/year regret. Five capabilities: (i) nonparametric bootstrap CIs for CVaR and the trimmed mean across the distribution families of M1; (ii) bias-variance audit of the Hill tail-index estimator with optimal threshold k/n; (iii) parametric-bootstrap bias correction for small-sample Weibull B-life; (iv) a mis-specification regret matrix on a TC1-like 16-channel MZI network with explicit USD/year cost; (v) sample-size planning curves giving the smallest n that achieves a 10% relative confidence interval on $\mathrm{CVaR}_{5\%}$ as a function of Pareto tail index.

M6 closes the toolkit's loop. Where M1–M5 *produce* risk numbers, M6 asks how trustworthy those numbers are and what they cost when wrong. Two practitioner audiences benefit immediately: engineers gain confidence intervals on every quantity that previously appeared as a single point estimate, and budget owners gain a dollar-translated regret for the modeling assumptions that quietly drive specification decisions.

14.1 Bias-variance decomposition of decision risk

For any risk-metric estimator $\hat{r}$ of an unknown population quantity r^*, the mean-squared decision error decomposes into

$$\mathbb{E}\Big[(\hat{r}-r^*)^2\Big] = \underbrace{\big(\mathbb{E}[\hat{r}]-r^*\big)^2}_{\text{bias}^2} + \underbrace{\mathrm{Var}[\hat{r}]}_{\text{estimator variance}} + \underbrace{C_{\text{mis}}^2}_{\text{mis-spec cost}}, \tag{12}$$

where the third term — absent from the textbook bias-variance treatment — is the systematic error introduced by fitting the *wrong distribution family* before the estimator is even run. The three terms are addressed respectively by Investigations 2–3 (bias and bootstrap correction), Investigation 1 (variance and nonparametric CIs), and Investigation 4 (mis-specification regret). Investigation 5 closes the budget loop by giving the sample size required to drive the variance term below a target tolerance.

14.2 Workflow

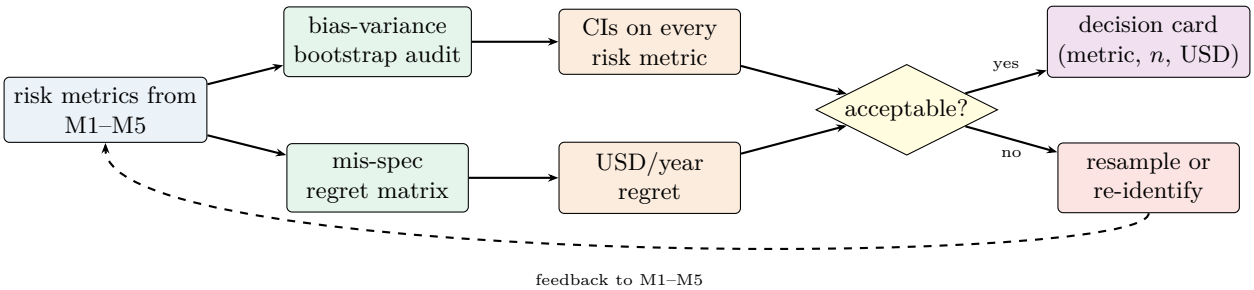


Figure 27: M6 workflow. Risk metrics produced by M1–M5 enter the meta-audit; the bias-variance branch returns confidence intervals on every metric; the mis-specification branch returns a USD/year regret matrix; an acceptance gate decides whether the result is admissible or whether the upstream module needs more samples or a different distribution family.

14.3 Bootstrap CIs on every risk metric

Investigation 1 generates $n = 1000$ samples from each of five distribution families (Gaussian, log-normal, Weibull, Pareto $\alpha = 2.5$, Pareto $\alpha = 1.8$) and runs 400 nonparametric bootstrap replicates of two competing tail statistics: $\mathrm{CVaR}_{5\%}$ and the symmetric 5%-trimmed mean.

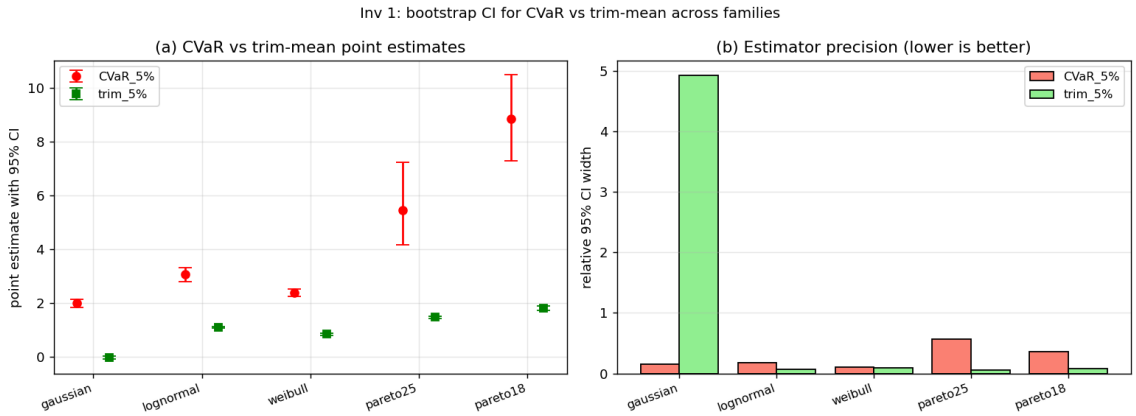


Figure 28: M6 Investigation 1: bootstrap 95% CIs and relative CI widths for $\mathrm{CVaR}_{5\%}$ and the symmetric 5%-trimmed mean across five distribution families. The CVaR estimator is precise for light-tailed families but its CI width grows by an order of magnitude on Pareto-tailed inputs; the trimmed mean remains stable across all families.

Bridge Equation

CVaR admissibility bridge. A risk metric should be *used* only when its bootstrap relative CI is below the decision tolerance the engineer is willing to absorb. For $\mathrm{CVaR}_{5\%}$ on Pareto inputs with $\alpha \leq 2$ this condition is unreachable at any practical sample size; the M2 trim-mean fallback is then mandatory rather than optional.

14.4 Hill threshold tuning and Weibull B-life correction

Investigation 2 sweeps the Hill threshold fraction k/n over $[0.02, 0.5]$ on Pareto samples with true $\alpha = 2.5$. The bias-squared and variance terms cross at a single k/n that minimizes the RMSE; this is the production-grade default that M1 should adopt rather than the textbook "upper decile."

Investigation 3 quantifies the small-sample bias of Weibull B10 estimated by maximum likelihood and shows that a single parametric-bootstrap pass with ~ 60 replicates removes most of the 10–20% low bias observed at $n = 10$, bringing the corrected estimate within a few percent of the truth at the sample sizes typical of qualification builds.

14.5 Mis-specification regret matrix on the TC1 MZI model

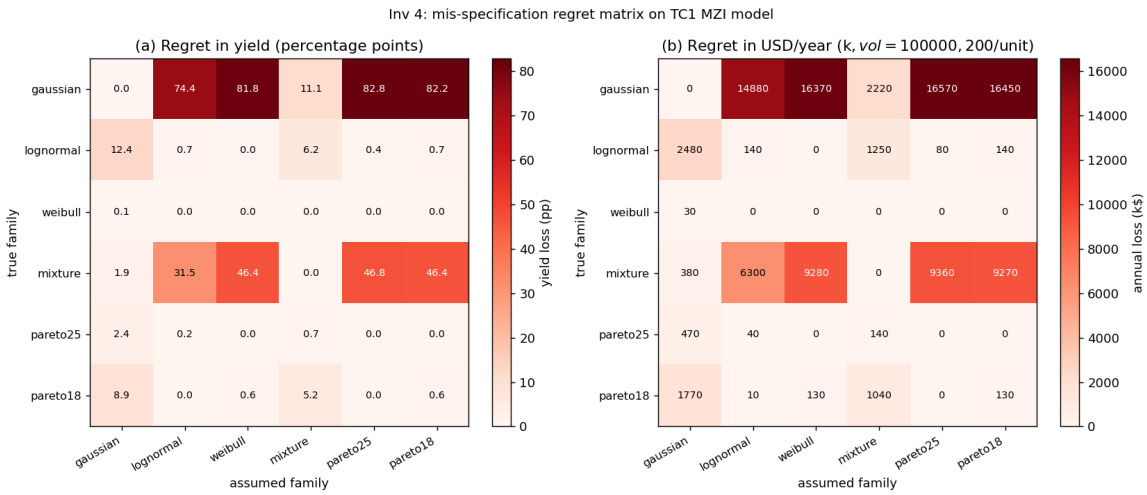


Figure 29: M6 Investigation 4: mis-specification regret matrix for the TC1-like 16-channel MZI network. Rows index the *true* phase-error distribution family; columns index the *assumed* family used to choose the pre-bias setpoint μ_b. Diagonal entries vanish; off-diagonal entries are the yield loss (left, percentage points) and the dollar regret (right, k\$/year at 10^5 units/year and \$200/unit).

The matrix monetizes a previously intangible quantity. Reading the row of any true family gives the cost of every wrong modeling assumption an engineer might make for that family; reading the column of any assumed family gives the worst-case cost of defaulting to that family across all possible truths. The minimax-safe assumed family is the one whose column maximum is smallest — a budget-defensible default in the absence of strong distributional evidence [28].

Design Rule

M6 design rule. Every distribution-aware decision shall report (i) the bootstrap relative CI on its driving metric, (ii) the worst-case mis-specification regret in USD/year for the candidate assumed families, and (iii) the smallest sample size that would have made (i) acceptable. A specification document that omits any of these three numbers is not yet production-grade.

14.6 Sample-size planning vs tail heaviness

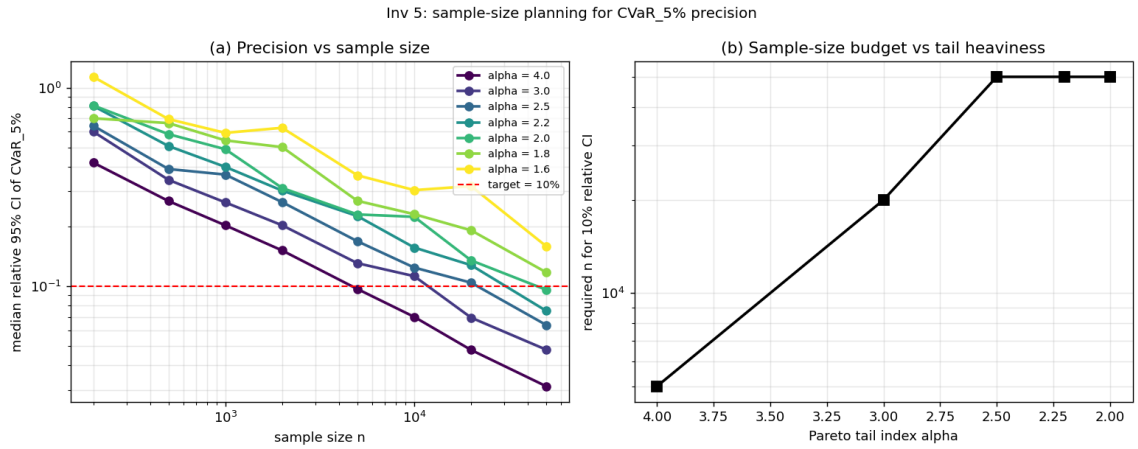


Figure 30: M6 Investigation 5: sample-size budget for a 10% relative CI on $\mathrm{CVaR}_{5\%}$ as a function of Pareto tail index α. The required n diverges as $\alpha \to 2^+$; for $\alpha < 2$ no finite sample size is sufficient, and the trim-mean fallback is mandatory.

Central Finding

M6 turns the bias-variance decomposition into engineering currency. Confidence intervals make every M1–M5 number auditable; the regret matrix monetizes the modeling choice; the sample-size curve identifies the regime in which the choice of *metric* matters more than the choice of *sample size*. The three together form the production-engineering quality bar for distribution-aware decisions.

14.7 M6 figures of merit

Table 9: M6 figures of merit.

FoM	Order of magnitude	Why
Relative bootstrap CI on $\mathrm{CVaR}_{5\%}$	10^{-1} (light tail) – 10^{1} (Pareto $\alpha < 2$)	Tells the engineer whether the metric is even measurable
Hill estimator RMSE at best k/n	10^{-1}	Irreducible accuracy floor for tail-index decisions
Weibull B10 MLE bias at $n = 10$	10^{-1} ($\sim$ 10–20%)	Drives whether warranty B-life specs need bootstrap correction
Worst-case yield regret	10^{0}–10^{1} percentage points	Quantifies the cost of one wrong family assumption
Worst-case dollar regret	10^{4}–10^{6} USD/year	Translates the modeling choice into budget language
n_{required} at $\alpha = 2.5$	10^{3}–10^{4} samples	Sample-size budget for tail-aware acceptance

15 Cross-Test-Case Comparison

Two cross-cuts of the test cases are useful. Table 10 contrasts the *Gaussian / traditional* treatment of each test case's headline figure of merit with the *distribution-aware* correction that C8 prescribes, and quantifies the gap. This is the table to read first: it answers the question "what does an engineer actually get wrong by defaulting to Gaussian?" Table 11 then summarizes the structural mapping across the four inevitabilities and the C8 module path used in each test case.

Table 10: Gaussian / traditional treatment vs. distribution-aware correction across the five C8 test cases. The "gap" column gives the order-of-magnitude consequence of using the traditional treatment when the underlying physics is non-Gaussian.

TC	**Headline FoM**	**Gaussian / traditional**	**Distribution-aware (C8)**	**Gap**
TC1	Channel / system yield	Cpk over $\delta\phi$ assumed Gaussian; symmetric $\pm 3\sigma$ window; nominal π setpoint	Quantile-Cpk on log-normal $\delta\phi$; empirical yield from MC; pre-bias Δx^* from CVaR optimum (M2)	+1 to +20 pp yield over-estimate
TC2	NETD ($\propto \sqrt{F}$)	Poisson statistics: $\mathrm{Var}(n) = \langle n \rangle$, $F = 1$; NETD $\propto \sqrt{\langle n \rangle}$	Bose–Einstein bunching: $\mathrm{Var}(n) = \langle n \rangle + \langle n \rangle^2$, $F = 1 + \langle n \rangle$; NETD inflated by $\sqrt{F}$	$\sqrt{F} \sim 1.05$–3 ($\sim$0.2–10 dB NETD penalty)
TC3	Single-photon certification	$g^{(2)}(0) < 1$ (second-order intensity correlation; classical statistics sufficient)	Wigner negativity volume $\mathcal{N}(\rho) > 0$; in-line homodyne tomography station (M5 T3)	False positives possible: $g^{(2)} < 1$ admits states with $\mathcal{N} = 0$
TC4	Coupling efficiency at converged setpoint	Nelder–Mead servo on quadratic cost surface; Gaussian readout noise assumed	Bayesian optimization with GP-LCB acquisition (M4 T2); two-distribution part / machine deconvolution	10–30% of nominal range (convergence bias under bimodal / heavy-tail noise)
TC5(a)	Rogue / peak-event statistics	Window standard deviation; 3σ event-rate prediction	Hill tail index $\hat{\alpha}$ on the upper decile; quantile / CCDF spec	Window-σ under-predicts events by 10–$10^3\times$
TC5(b)	Laser frequency noise	Linewidth as a single Gaussian standard deviation	Allan deviation $\sigma_y(\tau)$ on log–log axes; $1/f^\beta$ corner frequency	Order-of-magnitude σ_y error at long τ

Table 11: Cross-test-case summary across the four inevitabilities.

	TC1	**TC2**	**TC3**	**TC5**
Inevitability	I-2a	I-1	I-3	I-2b
Device anchor	16-ch CPO MZI	Mid-IR / TPV / THz	SPDC heralded source	Supercontinuum + laser noise
Distribution	Log-normal	Bose–Einstein	Wigner-negative	Pareto / $1/f^{\beta}$
Key FoM	quantile-Cpk, Δx^*	Fano factor, NETD ratio	negativity volume	Hill α, Allan slope
Hidden failure	Cpk yield overestimate	NETD underestimate	$g^{(2)}$ false positive	Window-std underestimate
C8 module path	M1 → M2 → M3	M1 → M3	M1 → M3 → M5	M1 → M3 → M5

TC4 (active alignment) is the cross-cutting test case and exercises modules M1, M4, and M5 simultaneously.

16 Conclusion

C8 closes a gap that the rest of the C-series cannot close from inside its own scope. C2 provides path-integral tolerance analysis, C3 designs subject to invariants, C4 verifies quantum hardware, C5 reconstructs from measurements, C6 integrates photonic circuits at the system level, and C7 validates the numerical pipeline. Each of these toolkits implicitly assumes that the statistical layer underneath its computations is Gaussian, and each is silently wrong in the regimes where modern photonics actually operates. C8 supplies the distribution-aware layer that the other six toolkits had to assume away, and it does so without requiring any of them to be rewritten: M2 re-uses C3's invariant catalogue as feasibility projections, M5's Wigner-tomography station re-uses C4's measurement primitives, and M6's regret matrix can be applied to any FoM produced by any C-series document.

Three claims summarize what the toolkit demonstrates. *First*, the four inevitabilities of non-Gaussianity are structurally distinct — fundamental quantum-statistical, compositional emergent, self-organized emergent, and quantum resource — and each requires a different mathematical response. There is no single "robust statistic" that handles all four, and any methodology that pretends otherwise will be wrong on at least one frontier. *Second*, the five pillars of any production-engineering program (design baseline, specification, assembly, inspection, capability reporting) each encode a hidden Gaussian assumption that, once named, can be replaced with a distribution-aware analogue without disrupting the workflow that sits on top of it. The replacements — pre-biased CVaR / trimmed-mean optima, quantile-Cpk, B-life, Fano-corrected NETD, Wigner negativity volume, Hill tail index, Allan deviation, and their counterparts in M3 — are not exotic; they are the standard objects of robust statistics, extreme-value theory, and quantum optics, deployed where the Gaussian default has been silently doing damage. *Third*, the meta-audit layer M6 turns every distribution-aware decision into a number with a confidence interval and a dollar cost, which is what a budget owner needs in order to authorize a methodology change.

Central Finding

The production-engineering toolkit of modern optical engineering must be built on a *distributional ladder* of identification, fitting, robust optimization, and quantile / extreme-value specification, with Gaussian as one node on the ladder rather than as the default. C8 makes this ladder concrete: six modules, five test cases, fifty-seven investigations, and a meta-audit layer that prices the cost of getting the distribution wrong. A specification document that omits (i) the bootstrap relative confidence interval on its driving metric, (ii) the worst-case mis-specification regret in USD/year for the candidate distribution families, and (iii) the smallest sample size that would have made (i) acceptable, is not yet production-grade.

The natural next steps are three: (a) experimental field validation of the synthetic-data results in a real CPO line and a real QKD source line, deferred to C8 future version; (b) back-port of M6's audit pattern to the headline figures of merit of C4 and C7, where the underlying quantities are estimated (not deterministic) and the same confidence-interval discipline applies; (c) development of contractual / specification language that lets the distribution-aware metrics travel across the foundry / customer / supplier interface, which is the adoption bottleneck flagged in Section 17. None of these is in scope for C8 v1, but each is now well-defined because of it.

17 What Is Not Shown

Honest Assessment

1. All ten simulation scripts use synthetic data with seed 42. C8's current version establishes *simulation validation*, not *experimental validation*.
2. Foundry-specific process statistics are not encoded.
3. Distributionally robust optimization is implemented at the level of a finite Wasserstein-1 ball via shifted-mean scenarios.
4. The Wigner inspection station of M5 is provided as a reference design only.
5. Quantum state tomography of non-diagonal states is delegated to C4 module M5.
6. Contractual / legal language for distribution-aware specifications is acknowledged as the next-stage adoption bottleneck but is out of scope for the technical document.

18 Reproducibility

All eleven simulation scripts (M1–M6 plus TC1–TC5, of which M1, M2, M3, M5 and M6 are at v1 and TC1–TC5 are at v0) follow the GAI file convention with timestamped root directories, `sim_config.json`, per-investigation subdirectories, `highlights.txt` files, and a root `summary_report.md`. All scripts use the deterministic seed 42, the matplotlib Agg backend, ASCII-safe encoding compatible with Windows CP950 / CP1252, and have no cross-script imports. The figure-path layer in this manuscript is implemented via one `\edef\detokenize` macro per script, defined at the top of the source file.

Table 12: C8 current version simulation script registry.

Script	Version	Macro	Investigations
`C8_M1_distribution_identification`	v1	`\MoneRoot`	6 distribution families
`C8_M2_robust_dro_optimization`	v1	`\MtwoRoot`	6 settings + bootstrap stability
`C8_M3_nongaussian_capability`	v1	`\MthreeRoot`	6 families + 1/f noise
`C8_M4_assembly_control`	v0	`\MfourRoot`	5 (alignment + deconvolution)
`C8_M5_inspection_gauge`	v1	`\MfiveRoot`	5 (incl. SOC avalanches)
`C8_M6_risk_quality_mispec`	v1	`\MsixRoot`	5 (bootstrap CI, Hill, B-life, regret, planning)
`C8_TC1_mzi_yield`	v0	`\TConeRoot`	5 (16-ch CPO MZI)
`C8_TC2_thermal_netd`	v0	`\TCtwoRoot`	5 (IR/TPV/THz)
`C8_TC3_wigner_negativity`	v0	`\TCthreeRoot`	5 (heralded source)
`C8_TC4_alignment_machine_noise`	v0	`\TCfourRoot`	5 (NM vs BO)
`C8_TC5_self_organized_photonics`	v0	`\TCfiveRoot`	5 (rogue + 1/f)

Total: 57 investigations across 11 standalone scripts, covering the four inevitabilities and the five pillars completely, with the M6 meta-layer auditing the statistical quality of every metric produced by M1–M5.

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